FRANCIS FAN

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EDUCATION

Johns Hopkins University Expected May 2027
M.S. Robotics; thesis in progress on model-based and imitation learning for surgical cautery

Union College June 2025
B.S. Mechanical Engineering; *cum laude*, Department Honors

RESEARCH EXPERIENCE

Surgical Robotics Researcher, IMERSE Lab, Johns Hopkins Oct. 2025–present
Advisor: Prof. Axel Krieger

Supervised autonomous bleeding control for splenic trauma (DARPA-funded)

- Built a monocular, vision-guided dVRK system for bimanual suction and cautery, integrating tool tracking, visual servoing, and collaborator-developed wound segmentation.
- Developed an EfficientNet tool–tissue contact detector; annotated ∼100,000 frames; achieved mean F1 of 0.738 in five-fold cross-validation with entire recordings held out.
- Built perfusion and suction loops and a custom tool adaptor; evaluated seven ex vivo AAST Grade I–II injuries, achieving 82.9% mean surface cauterization coverage.

Robotic vertebroplasty (ARPA-H-funded)

- Designed and tested a UR-mounted cannula-insertion end-effector delivering $>$160 N thrust in a $<$6 cm $\times$ 8 cm footprint.
- Designed a quasi-radiolucent modular backbone; ANSYS FEA predicted $<$0.1 mm tip deflection.

Researcher & Lab Manager, HEPIUS Innovation Lab, JHU Jan.–May 2026

- Fabricated focused-ultrasound transducers and characterized them using 3D hydrophone scans and in vitro models; managed lab operations for five researchers.

MANUSCRIPT UNDER REVIEW

Francis Fan, et al. "Supervised Autonomous Bimanual Suction and Cauterization for Bleeding Control in Low Grade Splenic Trauma." Submitted to ICRA 2027, Sept. 2026; under review.

SELECTED GRADUATE PROJECTS

CT-to-MRI Domain Adaptation Spring 2026

- Compared adversarial approaches to CT-to-MRI translation, including CycleGAN and StarGAN.
- Trained a ResNet34 U-Net with adversarial, L_1, and structural similarity losses; achieved SSIM of 0.865.

Intracranial Hemorrhage Detection Spring 2026

- Compared classical vision, SVM, and ResNet-18 pipelines on a dataset of 6,772 head CT slices.
- Engineered 19 image features for an RBF-kernel SVM that achieved 89.8% accuracy.

TECHNICAL SKILLS

Programming: Python, C++
Robotics & Control: ROS 2, Slicer-ROS, dVRK, daVinci Si, daVinci Classic, UR series robots, visual servoing
ML & Perception: PyTorch, OpenCV, PatchGAN, nnU-Net, EfficientNet, DeepLabCut, SAM 3, CVAT
Design & Simulation: SolidWorks, ANSYS FEA
Fabrication & Instrumentation: CNC and manual machining, 3D printing, cleanroom microfabrication, Arduino

ENGINEERING & LEADERSHIP

- **SAE Baja, drivetrain engineer and endurance driver (2023–2025):** designed and FEA-validated chain tensioners (FOS $>$5 at 500 N); completed 200+ race miles without mechanical failure.
- **Rocket Club, president (2022–2023):** led the design and build of a 3-foot rocket and managed the club budget.

PRESENTATIONS & AWARDS

- Oral presentation, ASME EFx, UNC Charlotte; **Technical Place Oral Competition Award** **2025**
- Oral presentation, American Chemical Society Spring National Meeting; ACS Presenter Recognition **2024**
- GlobalFoundries Summer Undergraduate Research Fellowship, Union College **2023**